

SET in Python

Introduction

A set in Python is a built-in data structure used to store a collection of unique elements. Sets are commonly used when duplicate values are not allowed and when mathematical set operations are required.

Definition of Set

A set is an unordered collection of unique elements enclosed within curly braces { }. Sets are mutable, but they do not support indexing or slicing.

Creating a Set

Example:

```
numbers = {1, 2, 3, 4}
```

Duplicate values are automatically removed:

```
data = {1, 2, 2, 3}
```

Characteristics of Set

- Unordered
- No duplicate elements
- Mutable
- Elements must be immutable
- Faster membership checking

Accessing Set Elements

Sets do not support indexing. Use loops to access elements.

```
for x in data:  
    print(x)
```

Adding Elements to a Set

```
numbers.add(5)  
numbers.update([6, 7])
```

Removing Elements from a Set

```
numbers.remove(3)  
numbers.discard(10)  
numbers.pop()
```

Common Set Methods

`add()`, `update()`, `remove()`, `discard()`, `pop()`, `clear()`, `copy()`

Common Set Methods

1. `add()`

```
s = {1, 2, 3}
s.add(4)
print(s)
```

2. `update()`

```
s = {1, 2}
s.update([3, 4, 5])
print(s)
```

3. `remove()`

```
s = {10, 20, 30}
s.remove(20)
print(s)
```

4. `discard()`

```
s = {10, 20, 30}
s.discard(40)
print(s)
```

5. `pop()`

```
s = {5, 10, 15}
s.pop()
print(s)
```

6. `clear()`

```
s = {1, 2, 3}
s.clear()
print(s)
```

7. `copy()`

```
s1 = {1, 2, 3}
```

```
s2 = s1.copy()
print(s2)
```

Set Operations

Union (`|`), Intersection (`&`), Difference (`-`), Symmetric Difference (`^`)

Python Set Operations – Explanation

1. Union (`|`)

Meaning:

The union of two sets contains all unique elements from both sets.

Syntax:

`A | B`

Example:

`A = {1, 2, 3}`

`B = {3, 4, 5}`

`print(A | B)`

Output:

`{1, 2, 3, 4, 5}`

2. Intersection (`&`)

Meaning:

The intersection of two sets contains only the elements common to both sets.

Syntax:

`A & B`

Example:

`print(A & B)`

Output:

`{3}`

3. Difference (`-`)

Meaning:

The difference of two sets contains elements present in the first set but not in the second set.

Syntax:

$A - B$

Example:

```
print(A - B)
```

Output:

{1, 2}

4. Symmetric Difference (^)

Meaning:

The symmetric difference contains elements present in either of the sets but not in both.

Syntax:

$A \wedge B$

Example:

```
print(A ^ B)
```

Output:

{1, 2, 4, 5}

Summary

Union (|): All elements

Intersection (&): Common elements

Difference (-): Elements in first set only

Symmetric Difference (^): Non-common elements

Python Set – Programming Examples

Set Operations

$A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

Union:

```
print(A | B)
```

Intersection:

```
print(A & B)
```

Difference:

```
print(A - B)
```

Symmetric Difference:

```
print(A ^ B)
```

Combined Practical Example

```
A = {10, 20, 30}
```

```
B = {20, 40, 50}
```

```
A.add(60)
```

```
B.update([70, 80])
```

```
print('Union:', A | B)
```

```
print('Intersection:', A & B)
```

```
print('Difference:', A - B)
```

```
print('Symmetric Difference:', A ^ B)
```

Applications of Set

- Removing duplicates
- Mathematical operations
- Fast searching